Lecture 29
Matched pairs; categories

Review

1 What does $df = \min(n_1 - 1, n_2 - 1)$ approximate?

the Welch df that software uses: a smaller df, so a conservative P-value

2 When do we pool for two means?

only when $\sigma_1 = \sigma_2$ is reasonable (similar SDs); unpooled is our default

3 What is the pooled SD s_p?

one SD from both samples: $s_p^2 = ((n_1 - 1)s_1^2 + (n_2 - 1)s_2^2)/(n_1 + n_2 - 2)$

The rank-sum test (non-parametric)

Wilcoxon rank-sum test

compares two independent samples without assuming normal populations

how it works

rank all $n_1 + n_2$ values together, then compare the two samples' rank sums

when to use it

small samples from skewed populations, outliers, or ranked (ordinal) data

more

Resources → Extras → Non-parametric Tests (also the signed-rank test for pairs)

Matched pairs: the idea

matched pairs

each value in one sample is paired with one in the other: same subject twice

the differences

$d = x_1 - x_2$ **for each pair: the n differences are the data**

the parameter

μ_d: **the mean difference in the population of pairs**

the big idea

a two-sample design becomes a one-sample t-test on the differences

The matched-pairs t-test

$H_0: \mu_d = 0$

no difference within pairs, on average

H_A

$\mu_d > 0$ **(upper),** $\mu_d < 0$ **(lower),** $\mu_d \neq 0$ **(two-sided)**

the test statistic

$t_{obs} = (\bar{d} - 0)/(s_d/\sqrt{n})$: $\bar{d}, s_d$ **= the mean and SD of the differences**

degrees of freedom

$df = n - 1$, **where n is the number of pairs**

assumptions

a random sample of pairs; the differences are normal, or $n \geq 30$

Data: hypnotism and pain

Subject	A	B	C	D	E	F	G	H
Before	6.6	6.5	9.0	10.3	11.3	8.1	6.3	11.6
After	6.8	2.4	7.4	8.5	8.1	6.1	3.4	2.0
d = after − before	0.2	−4.1	−1.6	−1.8	−3.2	−2.0	−2.9	−9.6

Randomly selected subjects had a sensory pain score measured before and after hypnotism; a lower score means less pain. X = a subject's score before, Y = the same subject's score after, $d = Y - X$. $n = 8$ pairs. (OpenStax, Example 10.11)

Example: does hypnotism reduce pain?

A study asked whether hypnotism reduces pain. Each subject's pain score was measured before and after hypnotism (lower means less pain), and the differences are normal. Are the scores lower after hypnotism, on average? (OpenStax, Example 10.11)

$n = 8$ **subjects;** d **= after – before;** $\bar{d} = -3.125,\ s_d = 2.91$**;** $\alpha = 0.05$

(a) Why matched pairs? State H_0 and H_A. Which tail?

(b) Find the SE and t_{obs}. What is the df?

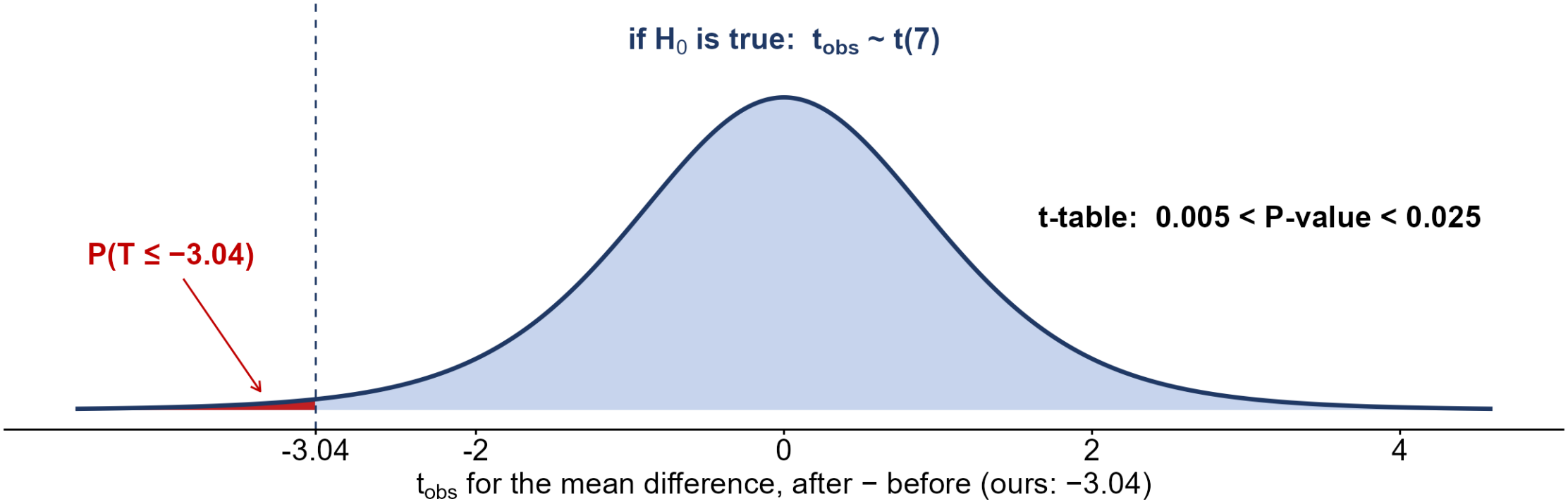
(c) Find the P-value from the t-table. What is the decision?

(a) the same 8 subjects, twice; $H_0: \mu_d = 0$ $H_A: \mu_d < 0$ **(lower-tailed)**

(b) $\text{SE} = 2.91/\sqrt{8} = 1.0288;\ t_{obs} = -3.125/1.0288 = -3.04;\ df = 8 - 1 = 7$

(c) $2.365 < 3.04 < 3.499$**:** $0.005 <$ **P-value** < 0.025**, below 0.05: reject** H_0

The hypnotism P-value



$0.005 <$ **P-value** < 0.025 **(software: 0.0094): pain scores are lower after hypnotism, on average**

Why pair?

Suppose we forgot the pairing and treated the hypnotism scores as two independent samples of 8.

before: $\bar{x} = 8.71,\ s = 2.18$; **after:** $\bar{x} = 5.59,\ s = 2.61$; $\bar{x}_{\text{after}} - \bar{x}_{\text{before}} = -3.125$

(a) Find the unpooled SE and t_{obs}.

(b) Compare with the matched-pairs test. Why is the paired SE smaller?

(a) $\text{SE} = \sqrt{2.18^2/8 + 2.61^2/8} = 1.2023;\ \ t_{obs} = -3.125/1.2023 = -2.60$

(b) paired $\text{SE} = 1.0288 < 1.2023$**: each subject is their own control**

Data: shot-put with each hand

Student	1	2	3	4	5	6	7
Dominant hand	30	26	34	17	19	26	20
Weaker hand	28	14	27	18	17	26	16
d = dominant − weaker	2	12	7	−1	2	0	4

Seven eighth graders pushed the shot-put with their dominant (writing) hand and with their weaker hand, in feet. X = the dominant-hand distance, Y = the weaker-hand distance, $d = X - Y$. $n = 7$ pairs. (OpenStax, Example 10.13)

Practice: shot-put with each hand

Seven eighth graders thought they could push the shot-put equal distances with either hand. Each pushed with their dominant hand and with their weaker hand; the differences are normal. Is there a mean difference between the two hands? (OpenStax, Example 10.13)

$n = 7$ **students;** d **= dominant – weaker (feet);** $\bar{d} = 3.71,\ s_d = 4.50;$ $\alpha = 0.05$

(a) Why matched pairs? State H_0 and H_A. Which tail?

(b) Find the SE and t_{obs}. What is the df?

(c) Find the P-value from the t-table. What is the decision?

(a) the same 7 students, both hands; $H_0: \mu_d = 0$ $H_A: \mu_d \neq 0$ **(two-sided)**

(b) $\text{SE} = 4.50/\sqrt{7} = 1.7008;$ $t_{obs} = 3.71/1.7008 = 2.18;$ $df = 7 - 1 = 6$

(c) $1.943 < 2.18 < 2.447:$ $0.05 <$ **P-value** < 0.10**, above 0.05: do not reject** H_0

Practice: which test?

1 **Mean nightly sleep: random samples of 40 first-years and 40 seniors**

two independent samples t-test: two separate samples, a quantitative response

2 **Reaction time of 25 drivers, rested and after a sleepless night**

matched-pairs t-test: the same 25 drivers, measured twice

3 **Share who exercise daily: random samples of 200 dog and 200 cat owners**

two-proportion z-test: two independent samples, a yes/no response

Summary: every test so far

test (samples)	test statistic	SE, unpooled	SE, pooled
one proportion z (one sample)	$z_{obs} = \frac{\hat{p} - p_0}{\text{SE}}$	$\sqrt{\frac{p_0(1-p_0)}{n}}$	–
one mean t (one sample)	$t_{obs} = \frac{\overline{x} - \mu_0}{\text{SE}}$	$\frac{s}{\sqrt{n}}$	–
two proportions z (independent)	$z_{obs} = \frac{(\hat{p}_1 - \hat{p}_2) - d_0}{\text{SE}}$	$\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$	$\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$
two means t (independent)	$t_{obs} = \frac{\overline{x}_1 - \overline{x}_2}{\text{SE}}$	$\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$	$s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$
matched pairs t (dependent)	$t_{obs} = \frac{\overline{d} - 0}{\text{SE}}$	$\frac{s_d}{\sqrt{n}}$	–

pool only for $H_0: p_1 = p_2$, or when $\sigma_1 = \sigma_2$ is reasonable. Non-parametric: the sign test (pairs), the rank-sum test (independent)

Data: teens and phones

Answer	Never	Rarely	Sometimes	Often	Total
Count	289	216	178	60	743
Proportion	0.39	0.29	0.24	0.08	1.00

A Pew Research Center survey asked 743 U.S. teens aged 13 to 17: "Are you losing focus in class by checking your cell phone?" (Week 2)

X = a teen's answer: Never, Rarely, Sometimes or Often. $n = 743$.

More than two categories?

So far a categorical response had two outcomes (yes or no), and we tested one p or $p_1 - p_2$. The Pew question has four answers, and someone claims all four are equally common among U.S. teens.

Never 289, Rarely 216, Sometimes 178, Often 60; $n = 743$

(a) What kind of variable is X? How many categories does it have?

(b) Can a test of one proportion p check the claim? Why or why not?

(a) categorical, with $k = 4$ categories

(b) no: one p covers one category; the claim is about all four, p_1, p_2, p_3, p_4, at once